

Recipe Sharing Platform with Meal Planner

Database Systems Engineering and Distributed Backend
Development

Y25 · 2026–2027 · Trimester 4

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Problem statement

- Recipes are often saved in scattered places.
- Weekly meal planning takes time and repeated effort.
- Ingredient lists are difficult to prepare across multiple recipes.

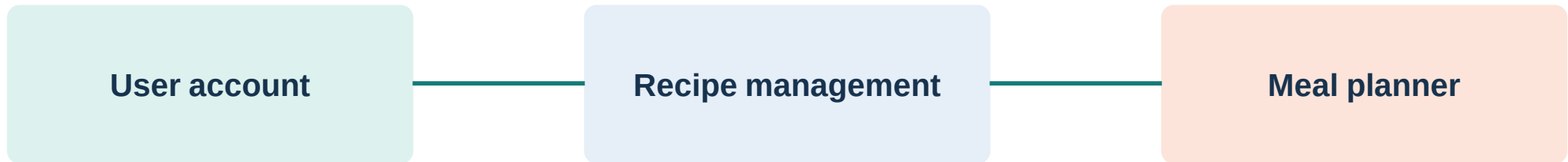
Need: one simple application for recipes, plans, and ingredients.

Objectives

- Provide accounts for creating, browsing, and sharing recipes.
- Support recipe search by category and recipe details.
- Organize selected recipes into a weekly meal plan.
- Generate a grocery list from planned recipe ingredients.
- Show nutrition information when available.

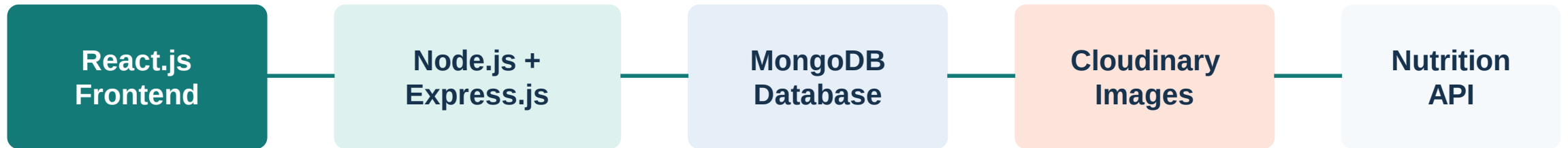
Proposed system

An integrated web platform for recipe discovery, meal planning, and grocery preparation.



Users can save and rate recipes for future use.

System architecture



The backend handles application requests and connects to storage and external services.

Modules and functionalities

- User registration and login.
- Recipe creation with ingredients and instructions.
- Recipe search, ratings, and saved recipes.
- Weekly meal planner.
- Grocery list generation.
- Nutrition information through an external API.

Frontend design

Home page

Recipe search
Categories
Featured recipes

Login / Registration

Account access
New user form

Recipe form

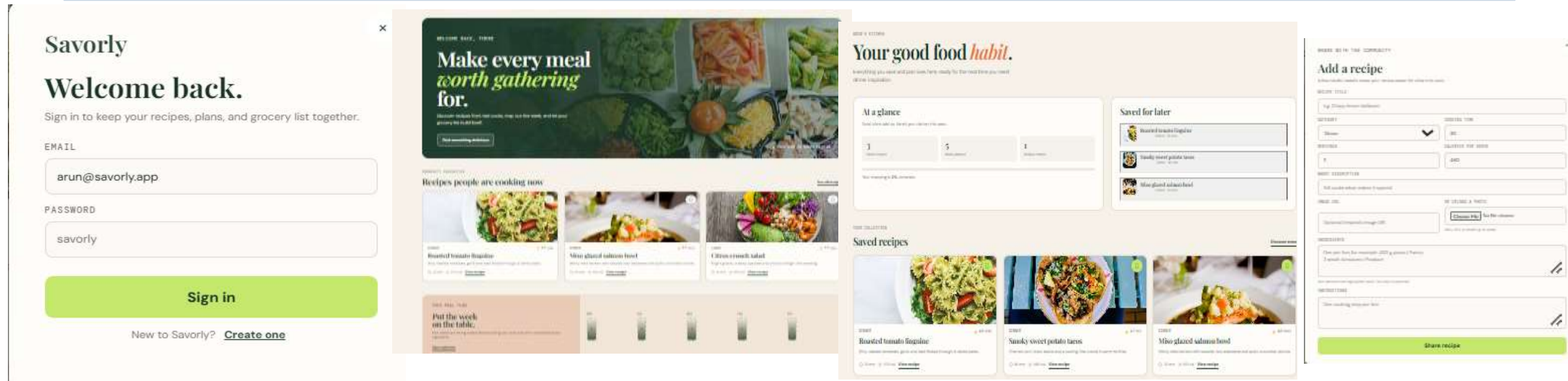
Ingredients
Instructions
Image upload

Dashboard

Saved recipes
Meal plan
Grocery list

Frontend implementation screenshots

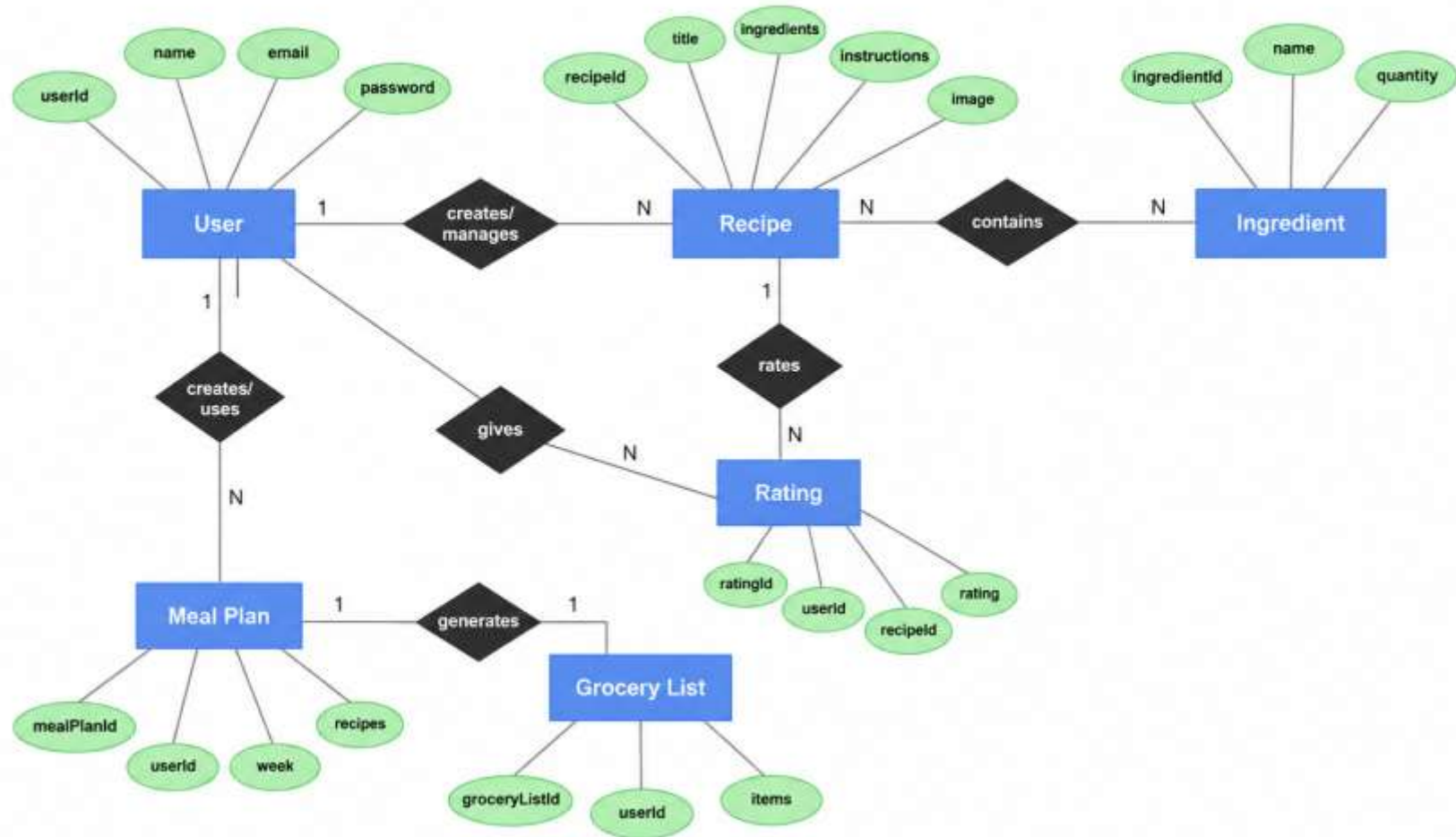
Inserting screenshots of the completed home page, forms, navigation, and dashboard.



Database design

- MongoDB stores user accounts, recipes, meal plans, and grocery lists.
- Cloudinary stores recipe images outside the database.
- The backend connects application requests to MongoDB and the Nutrition API.

ER DIAGRAM



Database tables and schema

Proposed collections and key fields

Users

name
email
password

Recipes

title
ingredients
instructions
category

MealPlans

userId
week
recipes

GroceryLists

userId
items

Sample database records

User

name: Arun
email: arun@example.com

Recipe

title: Vegetable Pasta
category: Dinner

MealPlan

week: 1
recipe: Vegetable Pasta

Current progress and challenges

- Project abstract submitted.
- Technology stack identified.
- Architecture and proposed data model prepared.
- Frontend screenshots are pending.
- Database tables and sample records are pending.
- Implementation evidence is required for Review 2.

Future work and conclusion

- Build the frontend pages and navigation.
- Implement the backend APIs and MongoDB collections.
- Add Cloudinary image uploads and Nutrition API integration.
- Capture evidence for the Review 2 demonstration.

The project aims to make recipe management, meal planning, and grocery preparation easier for users.